INFO 411 Assignment 2

Parkinson's Diagnosis Using Voice Signal Analysis Report

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Introduction

This report explores machine learning techniques in Julia to analyse various feature schemes, evaluate model performance, identify the most informative features for diagnosis, and contribute to the ongoing research on detecting of Parkinson's disease using a dataset obtained from Kaggle (Sakar et al., 2018).

Dataset Overview and Key Challenges

Upon first inspection several issues were identified. Firstly, the high dimensionality, with 754 attributes per entry, poses a risk of overfitting. Secondly, the dataset is imbalanced (188 patients and 64 healthy controls). Thirdly, each subject has three recordings, introducing repeated measures. Lastly, strong correlations between feature schemes which indicates feature redundancy.

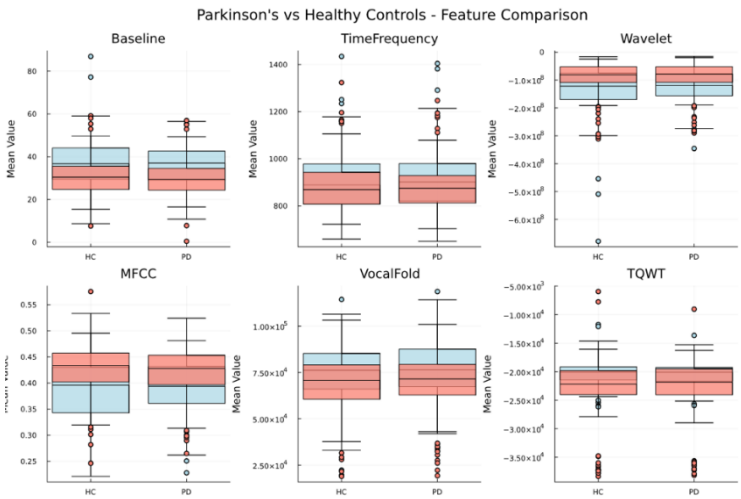


Figure 1. Overall feature comparison between Parkinson's and healthy controls by feature schemes

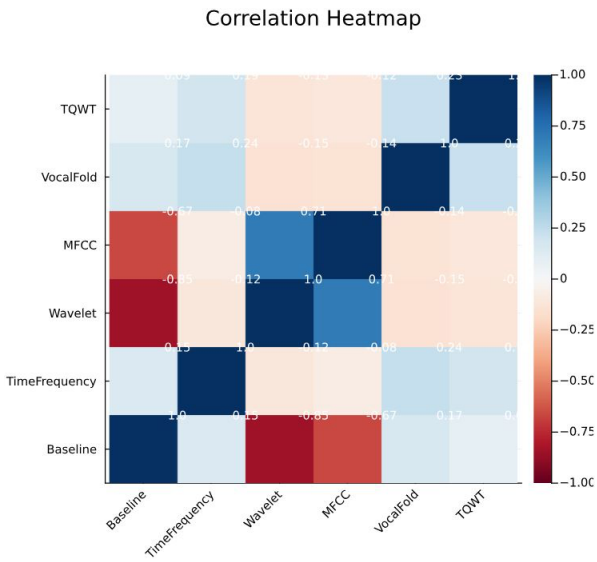


Figure 2. Correlation heatmap between feature schemes

The boxplots *(Figure 1)* illustrate general distribution differences between PD and HC, but large overlaps confirm that no single feature scheme can fully separate the two groups.

The heatmap *(Figure 2)* shows strong correlations, particularly between Wavelet and MFCC (r ~ 0.7) and between Baseline and Wavelet (r ~ -0.85). This confirms feature redundancy across schemes and supports the need for feature selection or dimensionality reduction.

Visual comparison and separability assessment of the feature schemes

To examine the overall data structure, PCA, t-SNE, and SOM were applied to assess separability between Parkinson's disease (PD) and healthy controls (HC) across six feature schemes. Because the feature ranges vary greatly across schemes, Z-score standardisation was essential. Without scaling, the visualisations produced distorted or inconsistent structures, for all three classifiers.

PCA *(Figure 3 - Left plots)* showed limited separation, as most schemes overlapped after standardisation, though MFCC and TQWT displayed slightly clearer boundaries. The PCA plots indicate that scaling reduces extreme variance and produces a more balanced projection, although class overlap remains.

t-SNE *(Figure 3 - Middle plots)* revealed non-linear manifolds and clearer local clusters after Z-score normalisation, confirming that scaling improves structure preservation, though overall separation remains weak.

SOM *(Figure 3 - Right plots)* visualisations further demonstrated that scaling strongly affects topology and interpretability, with MFCC producing the smoothest and most consistent U-Matrix.

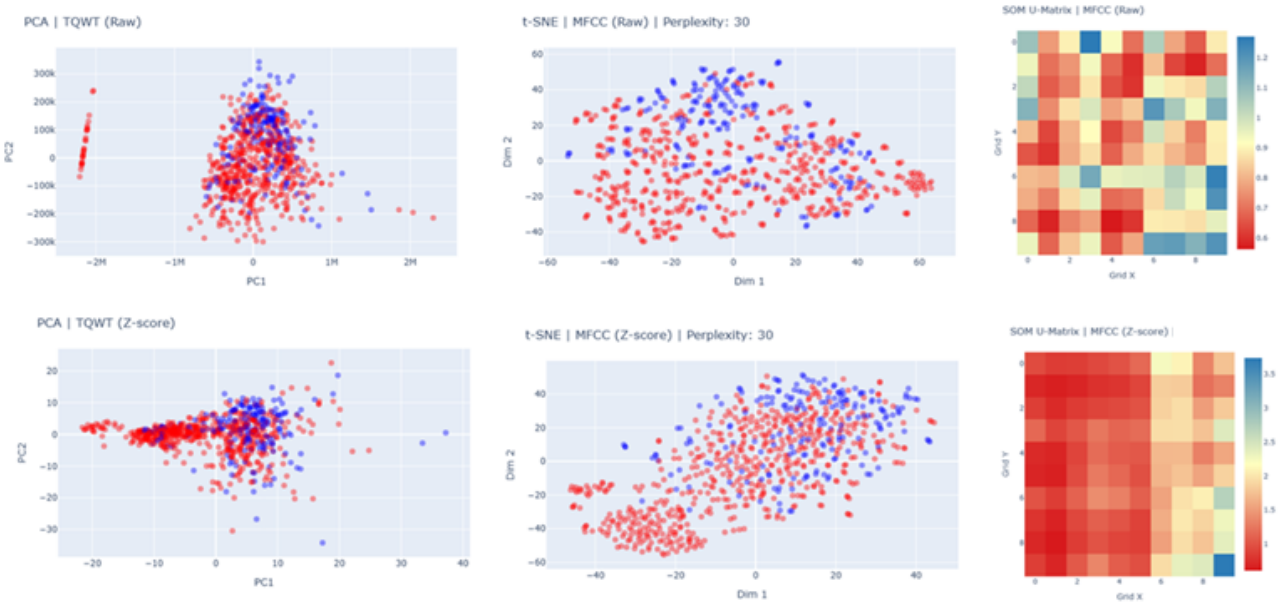


Figure 3 Visualisation Summary

Overall, MFCC and TQWT exhibited the best PD–HC separability, while TimeFrequency and VocalFold remained highly overlapped. These findings give an overall idea of the data structure and confirm that proper feature scaling and selection are crucial preparatory steps before model training and performance evaluation.

Feature scheme classification evaluation and comparison

To evaluate the discriminant ability of the feature schemes while avoiding data leakage, a grouped cross-validation mechanism was established so that no subject appeared in both training and testing folds. Building on the visual analysis, which suggested that MFCC and TQWT features offered the clearest PD–HC separability, here we quantitatively assessed their classification performance using three algorithms, SVM, Random Forest, and KNN. 5-fold cross-validation grouped by subject ID was applied across all six feature schemes. MFCC and TQWT consistently ranked highest across all classifiers, with peak accuracies around 0.80–0.82

depending on model, confirming that frequency-domain features capture Parkinson's-related voice patterns most effectively.

Random Forest showed the most stable and accurate performance with narrow fold-to-fold variation, whereas KNN and SVM displayed higher variability. The violin plots *(Figure 4)* highlight these trends where Random Forest produced compact accuracy distributions, KNN was more sensitive to data composition, particularly in TimeFrequency and Wavelet, and SVM yielded moderate but less differentiated outcomes.

The summary bar chart *(Figure 4)* further shows MFCC achieving the top mean accuracy followed by TQWT, while Baseline and VocalFold performed weakest. These findings demonstrate that ensemble methods better capture complex non-linear acoustic relationships and establish TQWT and MFCC with Random Forest as a strong baseline for subsequent feature-selection and model-tuning analyses.

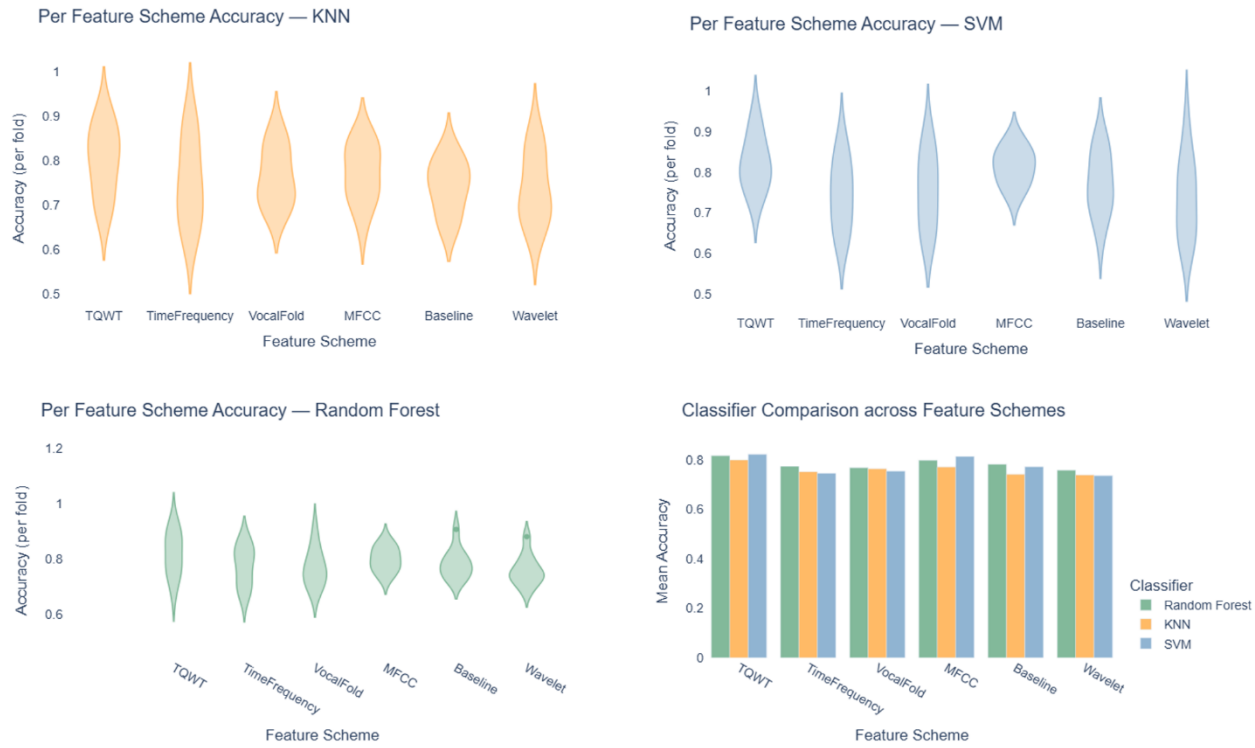


Figure 4. Feature Scheme Classification Evaluation Summary

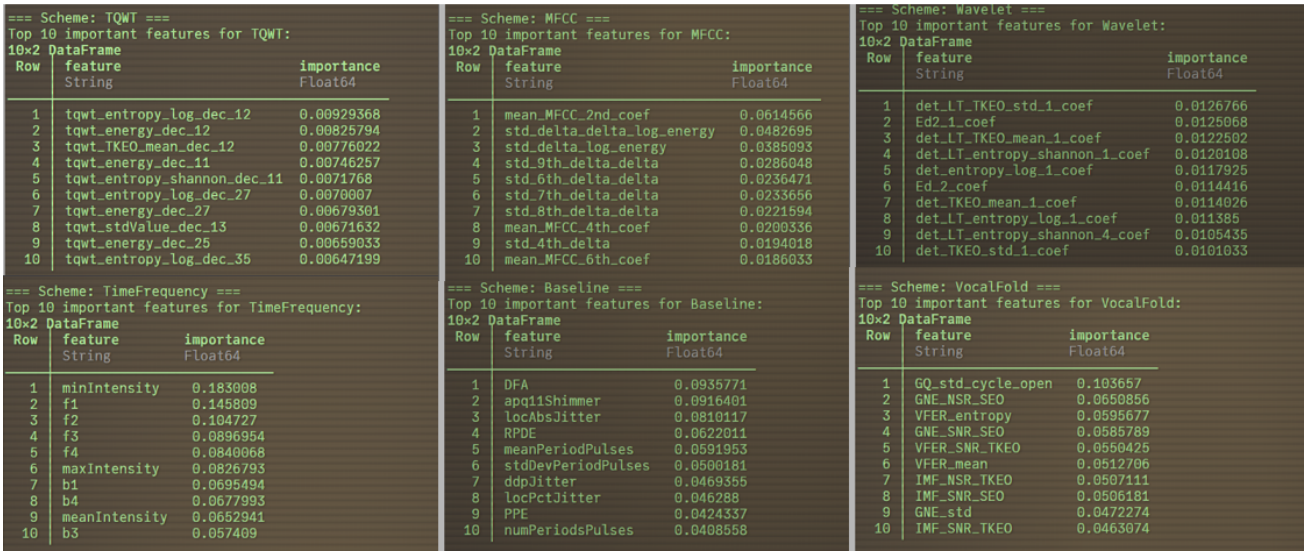
Feature selection & performance measures

To refine model efficiency and interpretability, two feature-selection methods, Random Forest feature importance and a correlation-based filter, were applied. Both RF and KNN classifiers were re-evaluated using accuracy, balanced accuracy, AUC, and F1 score.

RF was selected as a complex ensemble model capable of capturing non-linear relationships and feature interactions, while KNN served as a simple distance-based baseline for comparison.

This contrast allowed evaluation of whether feature reduction benefits both complex and simple classifiers.

At first, feature importance was analysed within each feature *(Figure 5)*. However, this approach was found to be less meaningful because the schemes differed greatly in scale, variable types, and dimensionality. Smaller schemes like TimeFrequency made almost every variable appear important, while larger ones like TQWT, MFCC diluted individual scores.



=== Scheme: TQWT ===
Top 10 important features for TQWT:
10×2 DataFrame

Row	feature (String)	importance (Float64)
1	tqwt_entropy_log_dec_12	0.00929368
2	tqwt_energy_dec_12	0.00825794
3	tqwt_TKEO_mean_dec_12	0.00776022
4	tqwt_energy_dec_11	0.00746257
5	tqwt_entropy_shannon_dec_11	0.0071768
6	tqwt_entropy_log_dec_27	0.0070007
7	tqwt_energy_dec_27	0.00679301
8	tqwt_stdValue_dec_13	0.00671632
9	tqwt_energy_dec_25	0.00659033
10	tqwt_entropy_log_dec_35	0.00647199

=== Scheme: MFCC ===
Top 10 important features for MFCC:
10×2 DataFrame

Row	feature (String)	importance (Float64)
1	mean_MFCC_2nd_coef	0.0614566
2	std_delta_delta_log_energy	0.0482695
3	std_delta_log_energy	0.0385093
4	std_9th_delta_delta	0.0286048
5	std_6th_delta_delta	0.0236471
6	std_7th_delta_delta	0.0233656
7	std_8th_delta_delta	0.0221594
8	mean_MFCC_4th_coef	0.0200336
9	std_4th_delta	0.0194018
10	mean_MFCC_6th_coef	0.0186033

=== Scheme: Wavelet ===
Top 10 important features for Wavelet:
10×2 DataFrame

Row	feature (String)	importance (Float64)
1	det_LT_TKEO_std_1_coef	0.0126766
2	Ed2_1_coef	0.0125068
3	det_LT_TKEO_mean_1_coef	0.0122502
4	det_LT_entropy_shannon_1_coef	0.0120108
5	det_entropy_log_1_coef	0.0117925
6	Ed_2_coef	0.0114416
7	det_TKEO_mean_1_coef	0.0114026
8	det_LT_entropy_log_1_coef	0.011385
9	det_LT_entropy_shannon_4_coef	0.0105435
10	det_TKEO_std_1_coef	0.0101033

=== Scheme: TimeFrequency ===
Top 10 important features for TimeFrequency:
10×2 DataFrame

Row	feature (String)	importance (Float64)
1	minIntensity	0.183008
2	f1	0.145809
3	f2	0.104727
4	f3	0.0896954
5	f4	0.0840068
6	maxIntensity	0.0826793
7	b1	0.0695494
8	b4	0.0677993
9	meanIntensity	0.0652941
10	b3	0.057409

=== Scheme: Baseline ===
Top 10 important features for Baseline:
10×2 DataFrame

Row	feature (String)	importance (Float64)
1	DFA	0.0935771
2	apq11Shimmer	0.0916401
3	locAbsJitter	0.0810117
4	RPDE	0.0622011
5	meanPeriodPulses	0.0591953
6	stdDevPeriodPulses	0.0500181
7	ddpJitter	0.0469355
8	locPctJitter	0.046288
9	PPE	0.0424337
10	numPeriodsPulses	0.0408558

=== Scheme: VocalFold ===
Top 10 important features for VocalFold:
10×2 DataFrame

Row	feature (String)	importance (Float64)
1	GQ_std_cycle_open	0.103657
2	GNE_NSR_SEO	0.0650856
3	VFER_entropy	0.0595677
4	GNE_SNR_SEO	0.0585789
5	VFER_SNR_TKEO	0.0550425
6	VFER_mean	0.0512706
7	IMF_NSR_TKEO	0.0507111
8	IMF_SNR_SEO	0.0506181
9	GNE_std	0.0472274
10	IMF_SNR_TKEO	0.0463074

Figure 5. Features importance by schemes

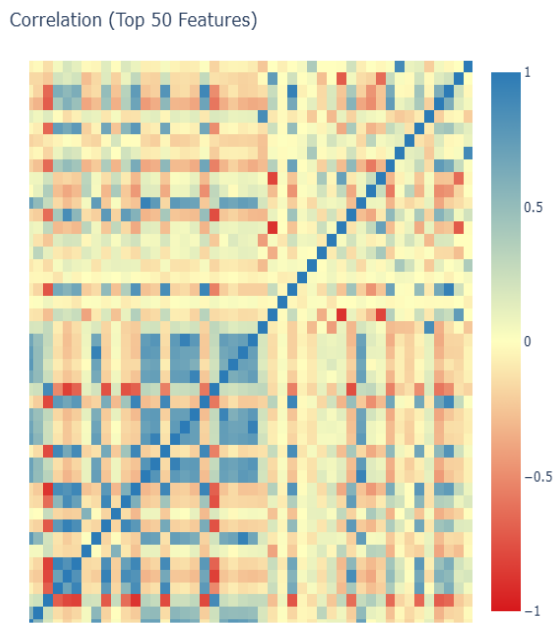


Figure 6. Top 50 Features Correlation Heat Map

To address this imbalance, the dataset with all feature schemes combined and global RF importance was re-evaluated to identify the most discriminative features for Parkinson's classification. Pearson's correlation filtering was then applied to remove redundant variables. From the top 50 important features (*Figure 6*), several strong correlations were found, so one feature from each highly correlated pair was removed, retaining the one with higher importance.

For Random Forest, thresholds between 0.6 and 0.9 were tested. Due to minor random variation, best performance fluctuates slightly across runs between thresholds 0.8 and 0.85. Therefore, 0.85 was chosen as the final threshold for further analysis. This reduced features from 754 to ~330 while maintaining accuracy ~0.82, balanced accuracy ~0.69, AUC ~0.85, and F-score ~0.88. KNN settled at the same threshold 0.85 and achieved its

best performance at K = 3 with accuracy ~0.80, balanced accuracy ~0.67, AUC ~0.76 and F-score ~0.87.

Both RF and KNN maintained stable accuracy at the 0.85 threshold, confirming that correlation-based feature reduction effectively removed redundancy while preserving predictive power, providing a compact and reliable baseline for further exploration.

Further explorations

After feature selection was performed on the full dataset, it was recognised that using all data for variable ranking could introduce data leakage. To ensure a fair and independent evaluation, a nested cross-validation framework was introduced. At the same time, the clear gap between accuracy (~0.82) and balanced accuracy (~0.69) suggested class imbalance, with more Parkinson's (PD) samples than Healthy Controls (HC). To address this, stratified cross-validation combined with class-balanced sampling was applied, ensuring equal class representation and preventing recordings from the same subject from appearing in both training and testing folds.

A 5×5 nested cross-validation design was adopted, where the inner loop performed hyperparameter tuning and the outer loop provided an unbiased evaluation. Z-score standardisation was applied only within the outer training data to avoid leakage from test folds. This setup provided a reliable and reproducible estimate of model generalisation.

At the selected correlation threshold of 0.85, both Random Forest and K-Nearest Neighbour were evaluated under the stratified nested framework. RF achieved an accuracy of ~0.76, balanced accuracy ~0.71, AUC ~0.82, and F-score ~0.83, showing strong and stable discrimination between PD and HC. KNN reached accuracy ~0.69, balanced accuracy ~0.67, AUC ~0.69, and F-score ~0.78, performing slightly lower but remaining consistent.

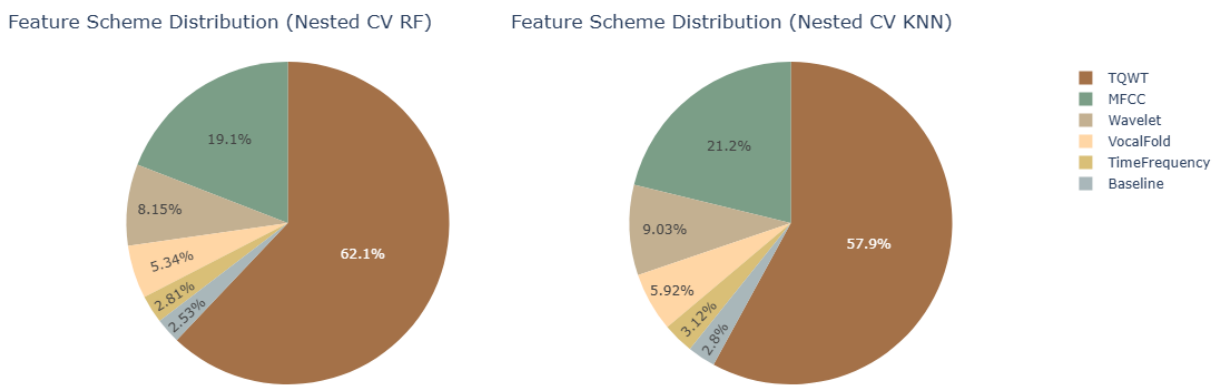


Figure 7. Feature Scheme Distribution Pie Chart

Feature selection patterns across all nested folds *(Figure 7)* showed that both classifiers concentrated on TQWT and MFCC features, though with slightly different proportions.

This consistent dominance confirms that time–frequency (TQWT) and spectral (MFCC) descriptors are the most informative for Parkinson's detection.

To explore possible gender effects, RF models were trained separately for male and female subsets. The male model reached both accuracy and balanced accuracy ~0.73 (AUC ~0.84, F1 ~0.78), while the female model showed accuracy ~0.76 but lower balanced accuracy ~0.68 due to data imbalance (~ 5 PD:1 HC).

Overall, introducing stratified nested cross-validation with class balancing effectively mitigated both data leakage and class imbalance, yielding reliable performance estimates.

Across all analyses, Random Forest remained the most consistent and accurate classifier, while TQWT and MFCC features consistently drove predictive strength, confirming their robustness across models and validation folds. Although gender-specific training slightly improved sensitivity in female subsets, the results also revealed that data imbalance remains the main limiting factor. Future work should collect age-stratified, gender-balanced datasets and investigate whether specific feature schemes (TQWT, MFCC) have differential diagnostic value across gender and age groups, potentially improving generalizability beyond single-gender approaches.

References:

Sakar, C.O., Serbes, G., Gunduz, A., Tunc, H.C., Nizam, H., Sakar, B.E., Tutuncu, M., Aydin, T., Isenkul, M.E. and Apaydin, H., 2018. A comparative analysis of speech signal processing algorithms for Parkinson's disease classification and the use of the tunable Q-factor wavelet transform. Applied Soft Computing, DOI: https://doi.org/10.1016/j.asoc.2018.10.022

Contributions: Our group focussed on collaborating for all tasks and findings together. Though, each group member was responsible for code quality and correct insight reporting for a defined section in each task.

	Aakanksha	**Any**	**Krupali**
EDA	Discussion	Discussion	Boxplots & Discussion
Visual comparison and separability assessment of the feature schemes	PCA	t-SNE	SOM
Feature scheme classification evaluation and comparison	SVM	RF	KNN
Feature selection & performance measures	Feature selection	Classifiers	Correlation
Further explorations	All		
Report	All		